
Experiment 3: Acceleration during Pendulum Motion

ME-330

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Week-02

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1 Learning objective

The objectives of this experiment are to provide the student with an opportunity to:

- Gain additional experience with Matlab.
- Create a simple program for data acquisition using toolbox and National Instruments USB-6001.
- Become familiar with breadboards, digital power supply, wiring, digital multi-meters, and other laboratory skills.
- Perform calibration of a MEMS accelerometer and create the calibration plot.
- Compare measured normal acceleration to acceleration calculated from angular velocity during fixed axis rotation.

2 Equipment checklist

- USB Data Acquisition Device (NI USB-6001).
- Pendulum with Potentiometer (10k Ω Bourns 6639s) pivot and end-mounted accelerometer (MMA7361LC).
- Needle nose pliers & wire strippers.
- Breadboard JE25
- Small flathead screwdriver.
- Digital multi-meter (DMM)
- Power supply and wire kit.

3 Introduction

During rotation about a fixed axis, the normal acceleration of a point can be calculated from the angular velocity of the rotation motion according to the following formula:

$$a_n = -\dot{\theta}^2 r \quad (1)$$

4 Laboratory instruction

This experiment will build on your work from the previous week. In addition to measuring the pendulum angle with a calibrated potentiometer, you will also measure normal acceleration of the end of the pendulum using a MEMS accelerometer. Schematic is shown in figure 1. Start by setting up the NI USB-6001 to acquire data from the potentiometer and accelerometer. Next, you will calibrate the y-axis of the accelerometer. Finally, we will compare measured normal acceleration of the pendulum end to normal acceleration calculated from the derivative of the pendulum angle.

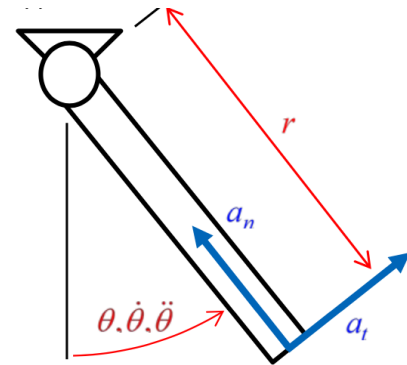


Figure 1: Schematic of the experimental setup.

5 Part-A

Creating data acquisition program using Matlab data acquisition toolbox

A sample Matlab program is provided in the appendix section of the instruction use that program as the starting point and modify accordingly. The modified Matlab data acquisition program should acquire data from two-channels of the NI-DAQ, display and record the data.

- Make sure that matlab program is acquiring data from two channels, The first channel should be acquiring potentiometer angle data and the second channel is acquiring the MEMS accelerometer data
 - For Input Range, use ± 10 Volts for the potentiometer and ± 10 Volts for the accelerometer.
 - Set the sample rate to 1000 samples/sec. We will be calculating a discrete derivative of the pendulum angle and a higher sample rate will reduce the noise from the process.
- The plot should be displaying the voltages from the accelerometer y-axis. You will need these for calibration of accelerometer.
- Please use the calibration equation from the previous laboratory experiment. If in doubt please contact the TAs for the correct calibration. The calibration should be able to convert the potentiometer output voltage to radians. Program this equation into the Matlab acquisition program. Verify that the calibration is producing the correct output by moving the pendulum to several known angles.

6 Part-B

Connect the y-axis accelerometer output to the data acquisition using a breadboard.

- For this experiment, a 3-axis accelerometer has been fixed to the end of the pendulum apparatus (see Figure 2). The accelerometer has been configured for $\pm 6g$ operation. The y-axis of the pendulum points towards the pendulum rotation axis (see Figure 3). For more information about the accelerometer, see [Pololu](#).
- Connect the accelerometer to the breadboard using the provided 3-wire cable.
- Connect the Vin and GND pins on the accelerometer to the 5 Volt supply power (already used to power the potentiometer).
- Connect the y-axis accelerometer output to AI1 (Differential Analog Input 1). **Need figure**
- Connect the DAQ GND, instead of the AI1-, to the ground on the breadboard. **Need figure**

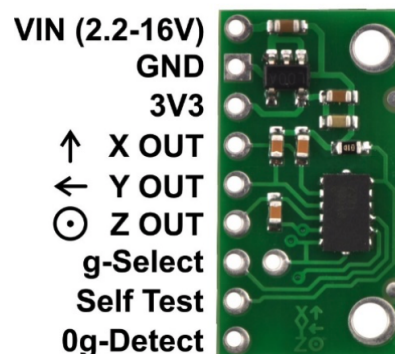


Figure 2: The MMA7361LC 3-axis accelerometer with voltage regulator.

7 Part-C

Run your acquisition program and acquire data from angular potentiometer and accelerometer

- When you have verified your circuit and connections, run your Matlab program.
- Modify the calibration bias as necessary to zero the pendulum as it hangs at the bottom.
- Confirm that you are acquiring data from the accelerometer. You will need the y-axis accelerometer voltages for the calibration step that follows (Part-D).

8 Part-D

Calibrate the y-axis accelerometer outputs.

- Complete the table below in your logbook by rotating the pendulum to the four shown positions and recording the output voltages on the y-axis of the accelerometer. Collect the voltage output from accelerometer for 20 seconds at a sampling rate of 20Hz. Make sure that you have saved these calibration file, if you need to verify your calibration. Fill the information in the table provided below.
- Plot the data points from the above table in order to create calibration equations for the y-axis acceleration measurements. Specifically, Plot ay_g vs. $ay_{v,avg}$ to find y-axis acceleration (in m/s^2) as a function of sensor voltage.

Table 1: Sample calibration data for experiment 2.

Pendulum orientation	y-axis acceleration ay_g (m/s^2)	Average output voltage $ay_{v,avg}$ (v)
Right	0.00	
Up	9.81	
Left	0.00	
Down	-9.81	

9 Part-E

Now add your calibration equation to acquisition program to display the pendulum angle in degrees during the data acquisition process.

- You will have to change the slope and intercept values in the provided Matlab program.
- Run your Matlab program without storing data and see if your program displays the correct angle.
- Show your program and results to the TAs to verify your program.

10 Part-F

Remove the gravity vector from your calibrated accelerometer output.

- In your logbook, determine the equation to remove the gravity vector from measured acceleration. That is, subtract acceleration due to gravity from the measured acceleration on the accelerometer y-axis as a function of potentiometer angle. The result will be the measured normal acceleration of the pendulum due to angular velocity.
- To compute the normal acceleration with gravity removed will required calculations with 2 signals. One approach is to modify the current plotandLog function file to do the required calculation.

11 Part-G

Create a time series plot of normal acceleration vs. time while the pendulum swings to rest from a horizontal starting position. Make sure to record data.

- Set the acquisition program to run for 20 second and collecting data at 1000Hz.
- Make sure the output file is named as “Expt_03_FirstName_Lastname.dat”
- Move the pendulum the horizontal position location.
- Start the acquisition program
- Release the pendulum
- Make sure the plot shows pendulum angular location and pendulum normal acceleration.
- Show the final output figure to the TAs.
- Open your output file and check the data.

12 Part-F

Plot the normal acceleration on a same plot 1) acquired from accelerometer and 2) calculated from pendulum angular location.

- Open your data file and read the angular position and accelerometer data.
- Calculated normal acceleration from the equation $a_n = \dot{\theta}^2 r$
- Plotting the noisiest data first (This way the other data will remain visible).

13 Part-G

Return lab space to prior condition

- Turn off the power supply.
- Remove wires connecting the power supply to the breadboard and return to the wall
- Remove all breadboard wires and place them back in the wire kit in an organized fashion.
- Remove the pendulum wires from the breadboard and set aside.
- Log off the computer.
- Make sure to copy the data to your hddrive.
- **Do this for every lab!**

14 Instruction

- Make sure to name the files your FirstName_ LastName_ QuestionNumber, use the format provided in the sample Matlab code. Make sure the submitted figure has the title. Include the apostrophe!
- Make sure to submit your post-lab assignment on the BBLearn website.
- Here are some details to help.
 - Plot the experimental data using red circles with MarkerSize 8. Experimental data points should always be plotted using markers without lines connecting the points
 - Plot curve from theory (or curve fit line) using a solid colored line with LineWidth 2.
 - Set the x and y-axis limits for each figures. Follow the instruction.
 - Make sure the major grid lines are visible
 - All plot text should be in Times font. You will need to specify this for the title, labels, and legends. The axis labels and numbers should be in 10-point font
 - The title should be in 10-point font
 - Make sure to add the legend. Keep in mind that legend should not hide the plotted data
 - For full credit make sure that your submitted files look similar to the sample figures shown in the document
 - If in doubt make sure to request the TAs for help

15 Post lab (50 points)

Postlab are due at Friday midnight. For this experiment submit the following items on BbLearn for your post-laboratory assessment.

1. Answer all question in the post-lab assessment on BbLearn

16 Appendix

Sample code for data acquisition using Matlab

```
1 % This is a genric program to acquire the data from NI- 6001
2 %% Sample data acquisition program
3 % Date: July 19th
4 % Dr. Vibhav Durgesh
5 % Rev 0.0
6 % This is basic program to acquire data from a data acquisition device.
7 % User has to provide aproprate information see — begining of the code
8 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
9 % THIS CODE OVERWRITE THE DATA. PLEASE MOVE THE FILES OR REMANE PRIOR TO
10 % RERUNNING THE CODE
11 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
12 clear all
13 close all
14 clc
15
16 %% ——— START USER REQUIRED INFORMATION ——— %%%%%%%%%%
17 outputFileName = 'Test.dat'; % This program will overwrite the file
18 slope = 1; % Slope for converting voltage to appropriate unit
19 intercept = 0; % Intercept for converting voltage to appropriate unit
20 Fs = 30; %Sampling rate Data per sec
21 T = 3; % Time to acquire data
22 choice = 0; %% Set to 1 to write data to a file otherwise 0
23 %% ———END USER REQUIRED INFORMATION ——— %%%%%%%%%%
24 d = daq.getDevices;
25 s = daq.createSession('ni');
26 addAnalogInputChannel(s,'Dev2','ai0','Voltage');
27 s.Rate = Fs; % Sampling rate modify as required
28 s.DurationInSeconds = T; % Sampling time modify as required
29 if choice == 1
30     fid1 = fopen(outputFileName,'w'); % File name to store your data
31     fprintf(fid1,'%s \t %s \n','Time(s)','Volt (v)');
32 else
33     fid1 = 0;
34 end
35 lh = addlistener(s,'DataAvailable',@(src, event)plotAndLogData(src, event,
36     fid1,slope,intercept,choice));
37 errorListener = addlistener(s,'ErrorOccurred',@(src,event) disp(getReport(
38     event.Error)));
39 drawnow
40 startBackground(s);
41 pause
42 delete(s)
43 delete(lh)
44 if choice == 1
45     fclose(fid1);
46 end
```

```

1 function plotAndLogData(src,event,fid,m,c,choice)
2 time= event.TimeStamps;
3 voltage=event.Data;
4 plot(time, voltage, 'k.', 'markersize',6)
5 title('Press any key when data acquistion is done','fontsize',14)
6 hold on
7 xlabel('time (s)')
8 ylabel('voltage (v)')
9 if choice == 1
10     for ii = 1:length(voltage)
11         fprintf(fid, '%3.5f \t %3.5f \n',time(ii),voltage(ii)*m + c);
12     end
13 end
14 end

```